

Physical Layer

This layer deals with Electrical Engineering stuff. It's how we understand what is actually happening on a wire.

1. Transmission Impairment

Attenuation, Distortion and Noise.

2. Transmission Media

Twisted Pair, Coaxial Cable and fiber optics.

3. Wireless Transmission

Radio wave, Satellite, Cellular Telephony.

4. Switching

Circuit switching, Packet switching and Message switching

Transmission Impairment

Attenuation: the loss of energy as the signal propagates.

Distortion: Different frequencies travel at different speed so the wave form spread out.

Noise: Unwanted energy combines with the signal.

Signal to noise ratio - This is measured in terms of the ratio of signal power to noise power. Usually this is measured in decibels, in terms of $10 \log_{10} S/N$. So an S/N of 100 = 20 dB.

Shannon's Capacity: maximum data rate of noisy channel given by

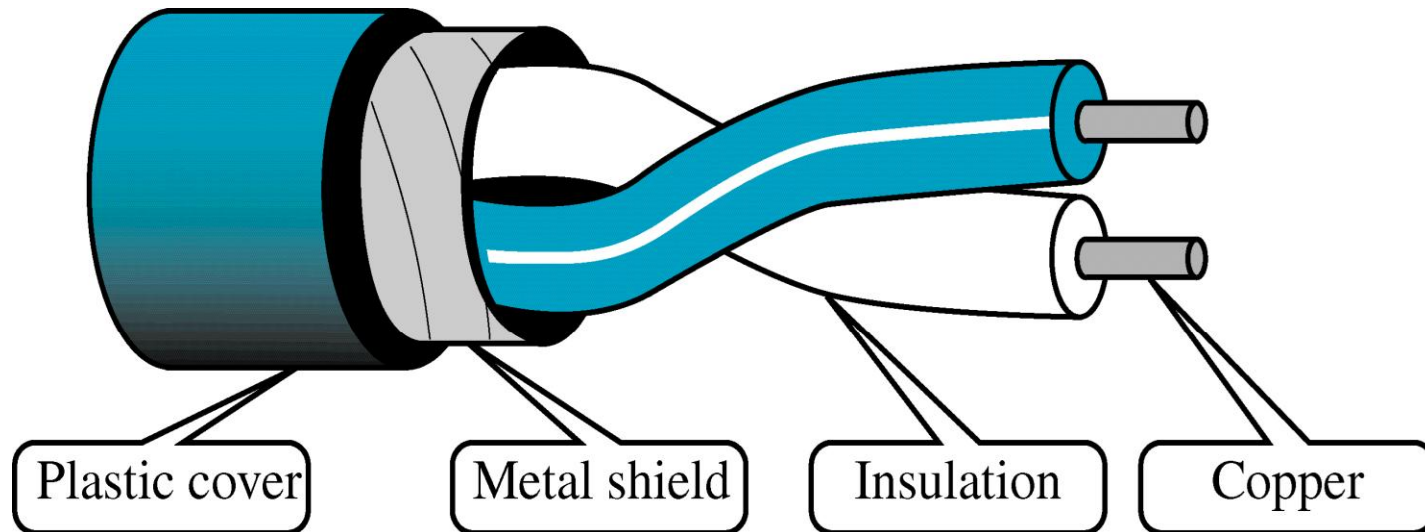
$$\text{maximum data rate} = B \log_2 (1 + S/N)$$

B – bandwidth of channel.

Transmission Media - Guided

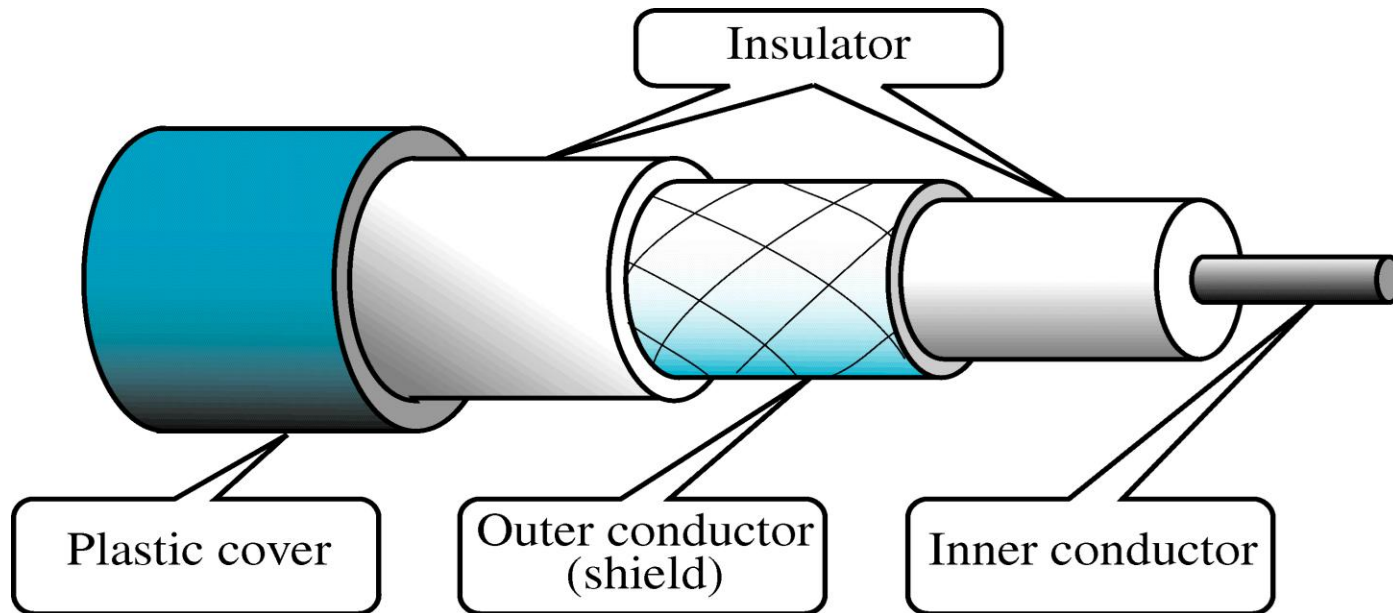
TWISTED PAIR:

- Simply two wires twisted together - the twisting cuts down on electrical interference.
- Heavily used in the phone system.
- Category 3 and 5 - with 5 having more twists and better insulation.



BASEBAND COAXIAL CABLE

- Used for digital transmissions (called **baseband**.)
- Good noise immunity.
- Data rates as high as 1 **Gbps** for short distances.
- Now being replaced by fiber.



BROADBAND COAXIAL CABLE:

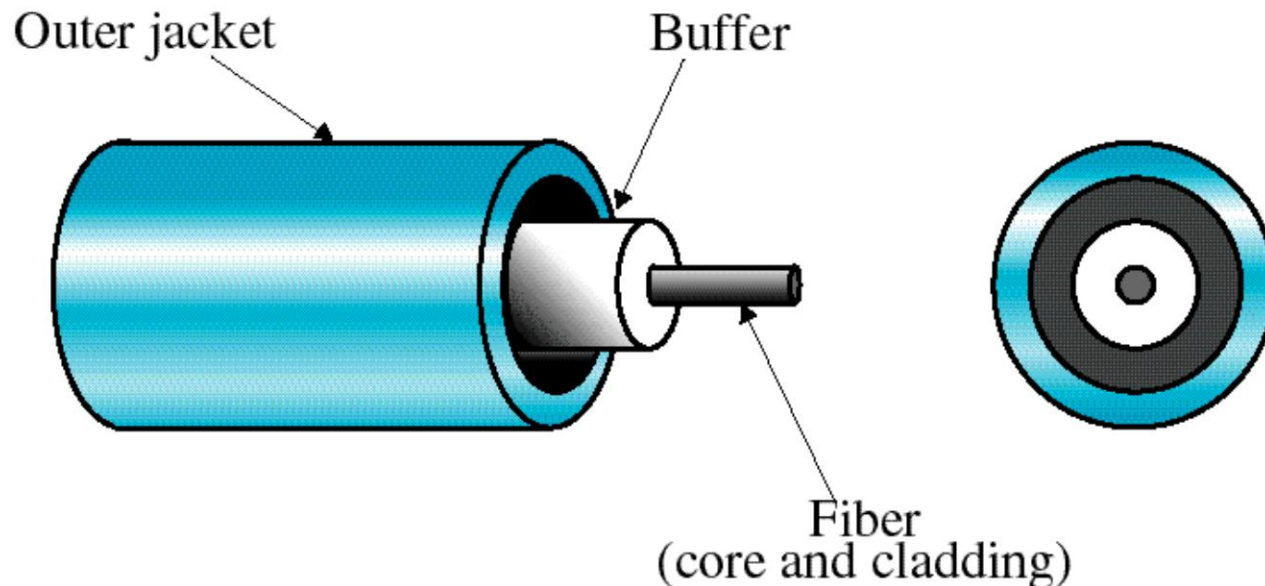
- Used for analog transmissions (called broadband.)
- Can run 300 MHz for long distances.
- Analog signaling has better S/N than digital signaling.
- Interfaces must convert digital signals to analog and vice versa.
- Designed for long distances - can use amplifiers.

FIBER OPTICS:

- Transmission of light through fiber - properties include total internal reflection and attenuation of particular frequencies.
- Fiber Optic Networks - can be used for LANs.

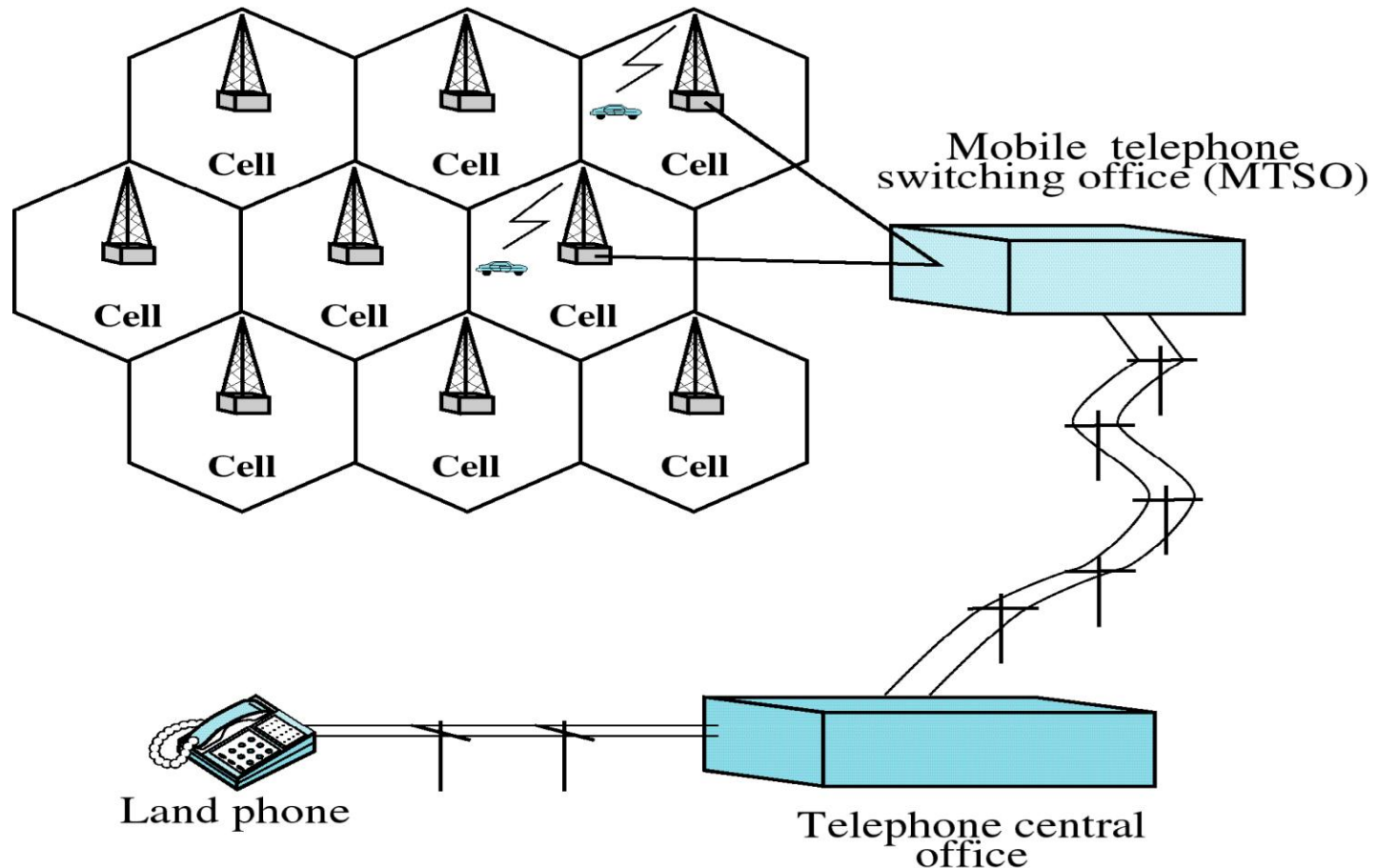
Comparison of Fiber Optics and Copper Wire

	Fiber	Copper
Bandwidth	Higher	Lower
Distance between repeaters	30 Km	5 Km
Interference	Low	High
Physical Flow	Smaller/Lighter Uni-directional	Bi-directional



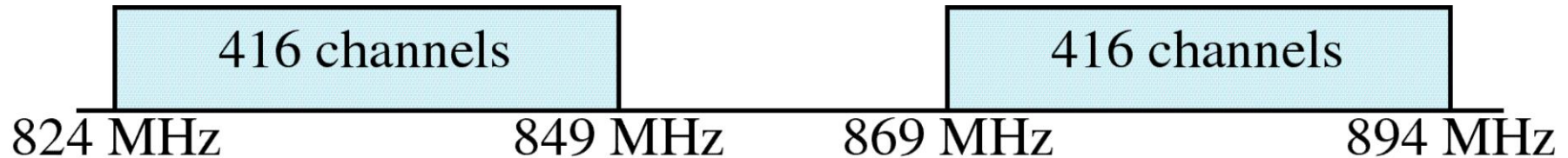
Cellular Telephony:

- Each cellular service area is divided into small regions called cells.
- Each cell contains an antenna and is controlled by a small office called the cell office.
- Each cell's office, in turn, is controlled by a switching office called a mobile telephone switching office (MTSO).
- The MTSO co-ordinates communication between all of the cell offices and the telephone control office.
- It is also responsible for connecting calls as well as recording call information and billing.



Cellular Bands

Cellular transmission is analog. FM is used for communication. Full-duplex 416 channels used for mobile carrier and another 416 channels used for landline carrier.



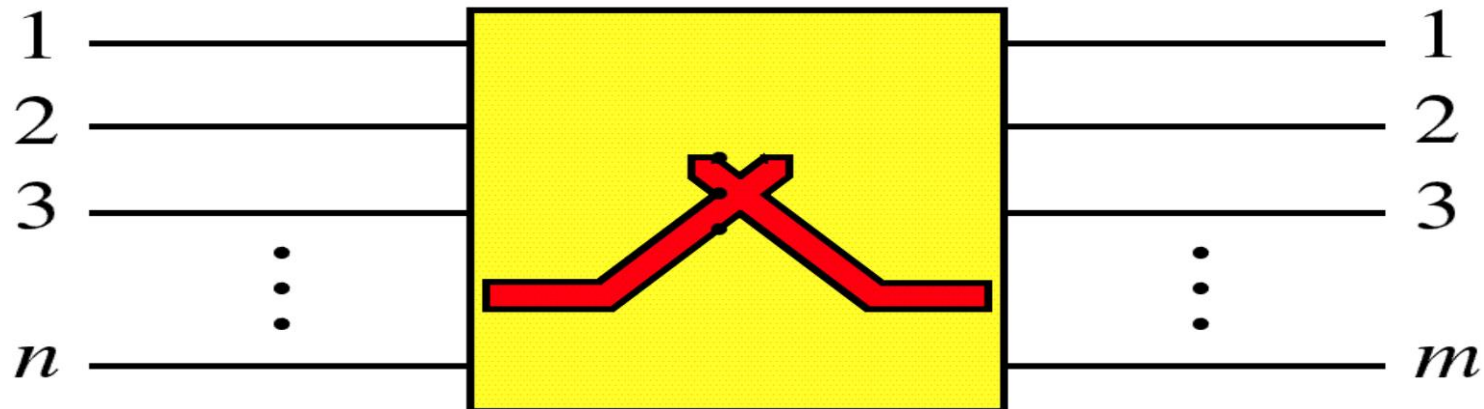
Transmitting: Mobile to landline phone

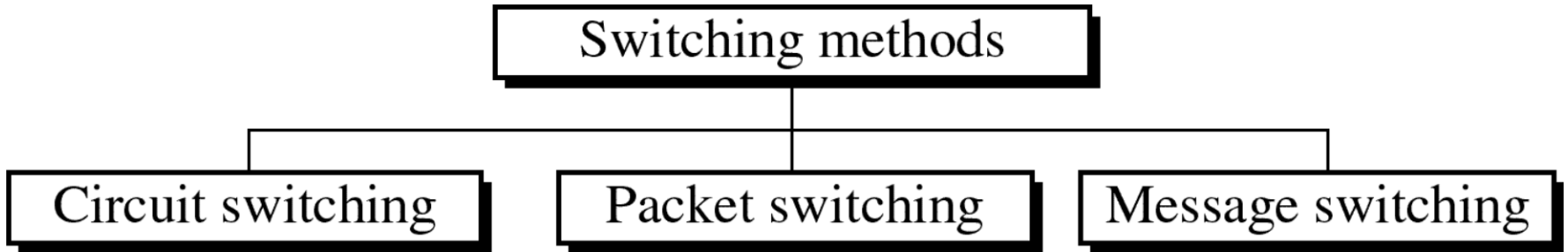
Receiving: Landline phone to mobile

Handoff: Mobile moves from one cell to another.

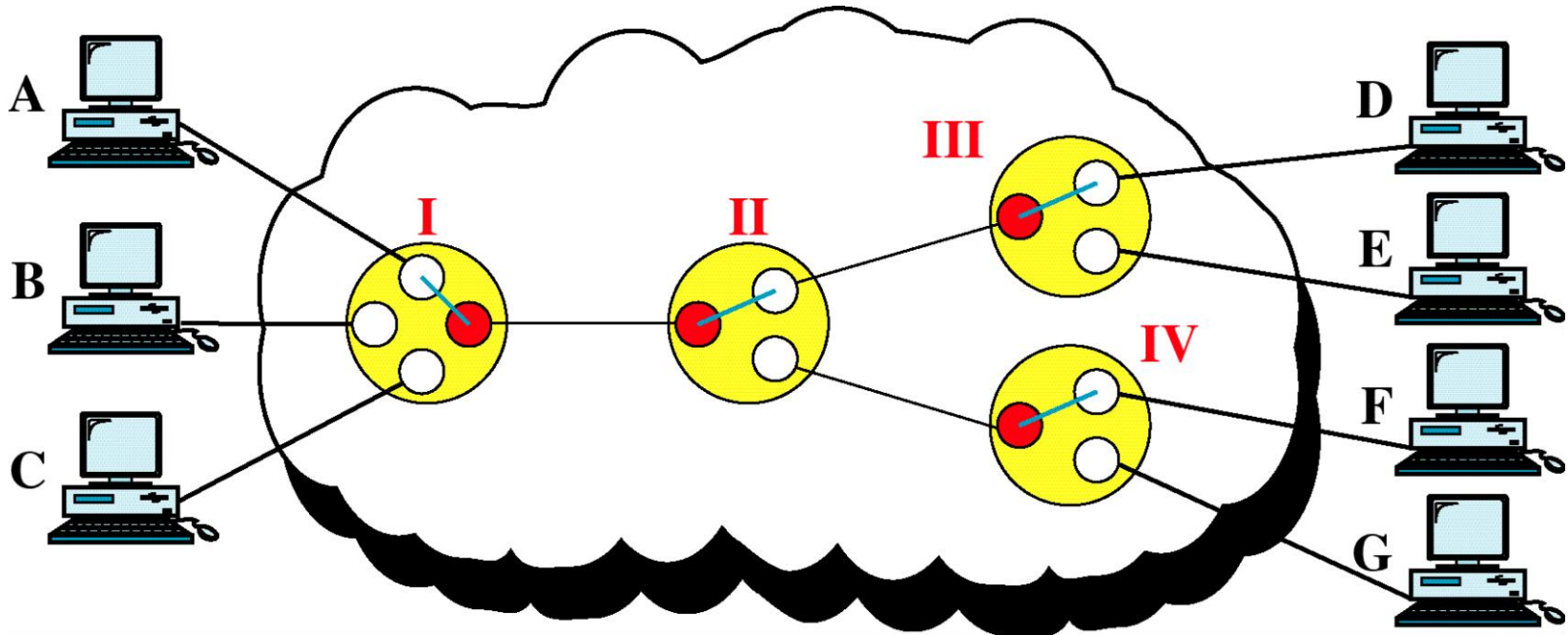
Switching

Switches are hardware and/or software devices capable of creating temporary connection between two or more devices linked to the switch but not each other.





Circuit-Switched Network

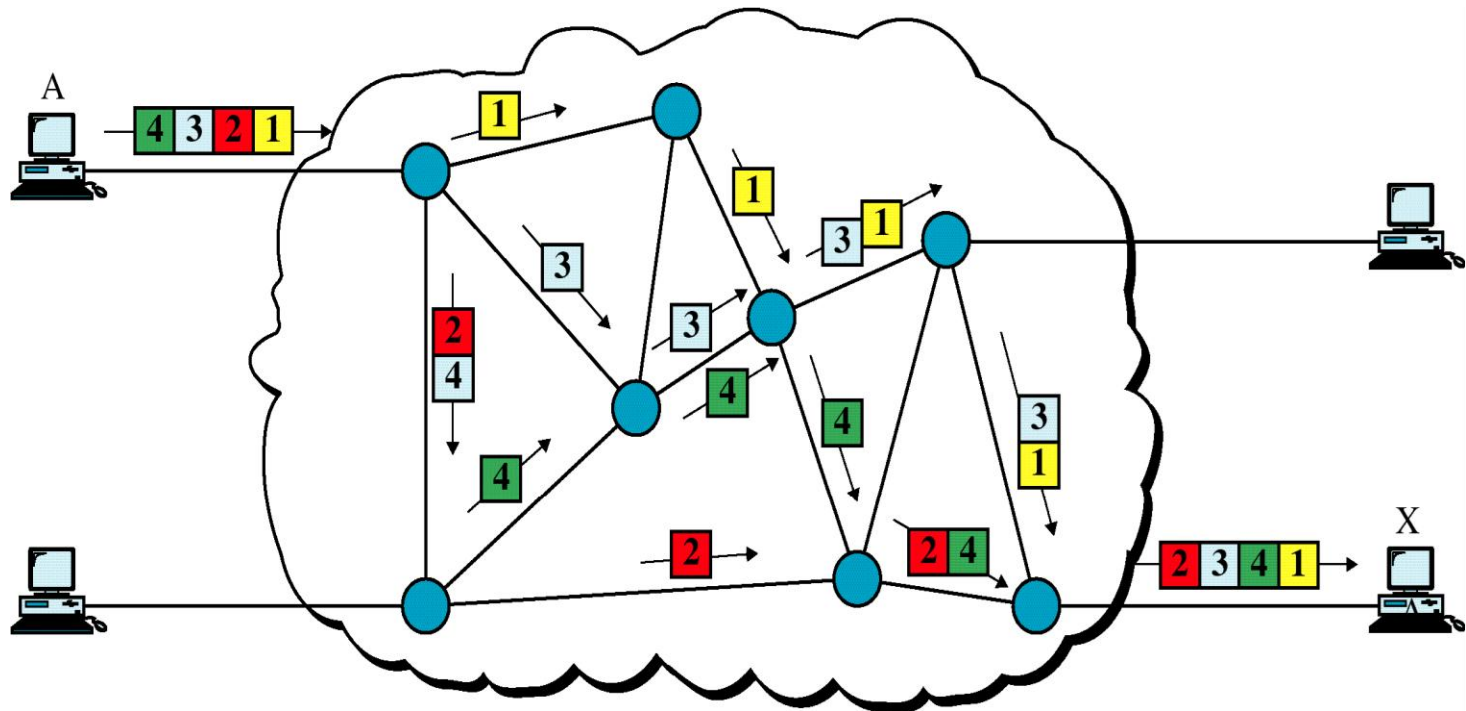


Packet switching

Datagram approach

Virtual circuit approach

Datagram Approach

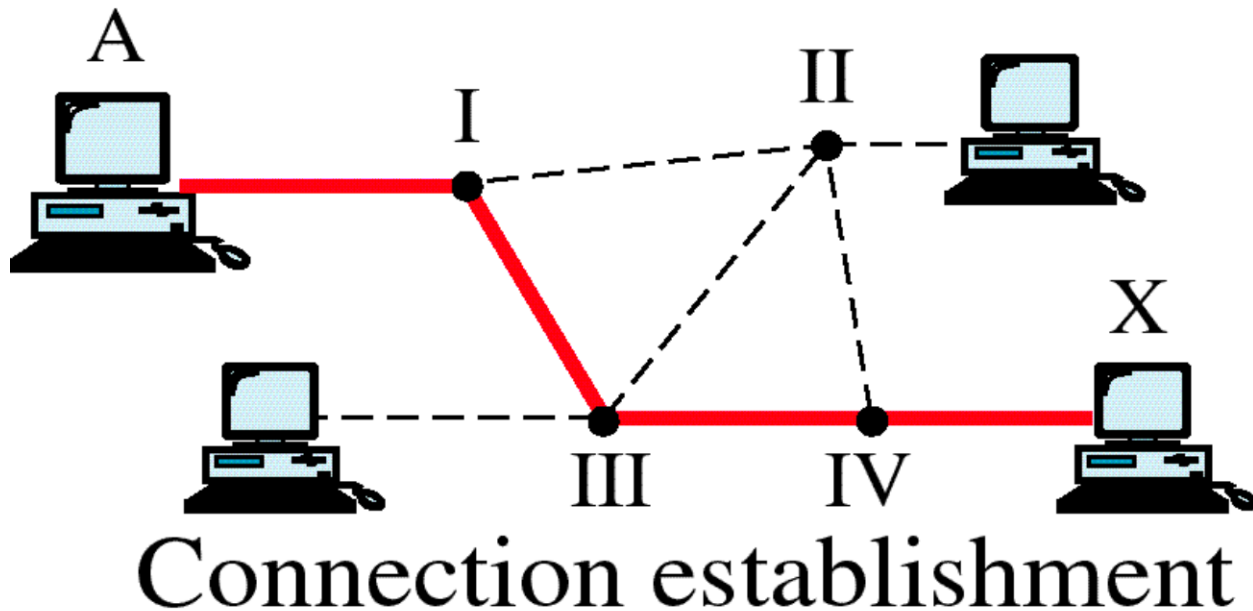


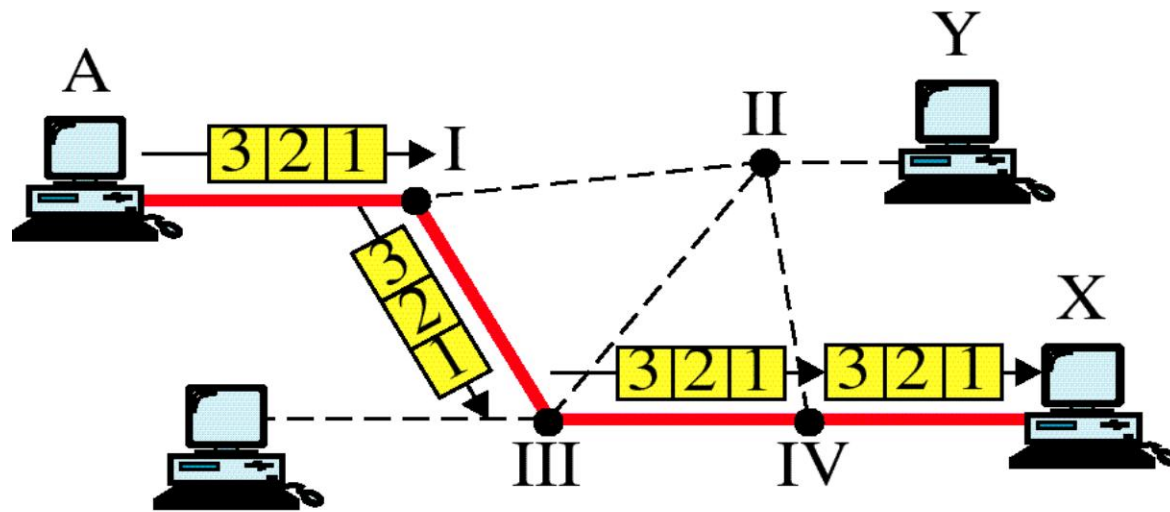
Virtual Circuit Approach

Virtual circuit transmission is implanted in two forms:

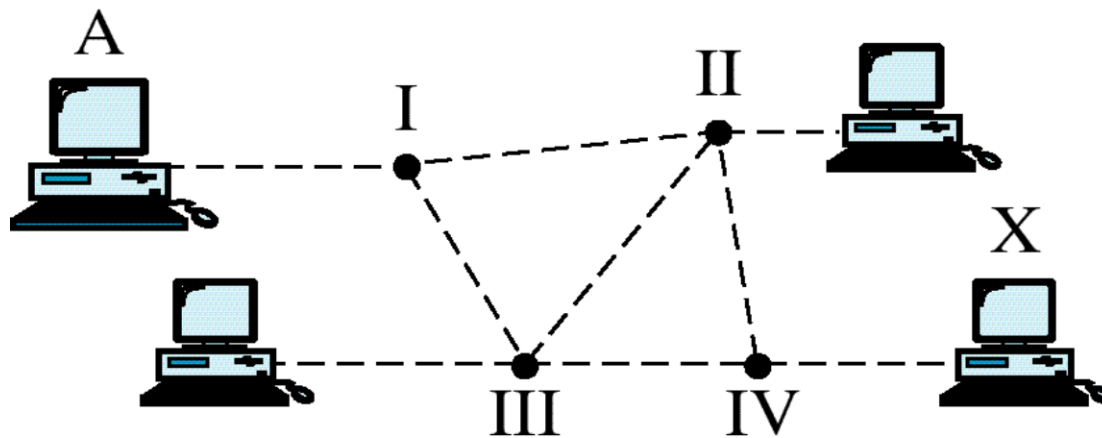
- Switched Virtual Circuit.
- Permanent Virtual Circuit.

Switched Virtual Circuit





Data transfer



Connection release

Permanent Virtual Circuit:

- Agreement between network and end users.
- Circuit is dedicated to the specific user.
- No one else can use it.
- It can be used without connection establishment and connection termination.

Message Switching

A node receives a message, store it until the appropriate route is free, then sends it along.

